

- 4 Adipocytes are cells found in adipose tissue in mammals. These cells absorb glycerol and fatty acids to make triglycerides for long-term storage.

Fig. 4.1a shows a glycerol molecule and three fatty acids. Fig. 4.1b shows the triglyceride molecule formed from these components.

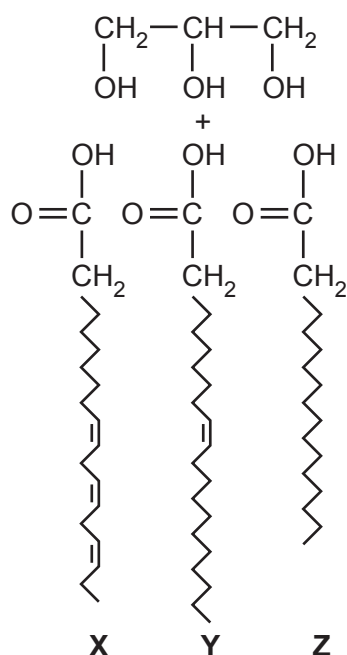


Fig. 4.1a

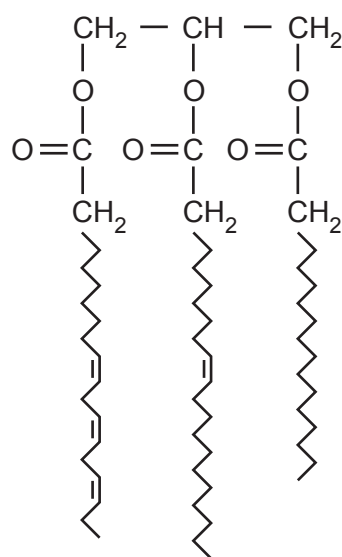


Fig. 4.1b

- (a) (i) State the name of the bonds that form between glycerol and fatty acids.
 [1]
- (ii) When a bond forms between glycerol and a fatty acid, water is a product of the reaction.
 State the name given to this type of reaction.
 [1]
- (iii) Describe the differences between the fatty acids, **X**, **Y** and **Z**, shown in Fig. 4.1a.

 [3]

(b) (i) State reasons why triglycerides are described as hydrophobic.

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(ii) Explain why triglycerides are **not** suitable as a component of cell surface membranes.

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..... [2]

[Total: 9]